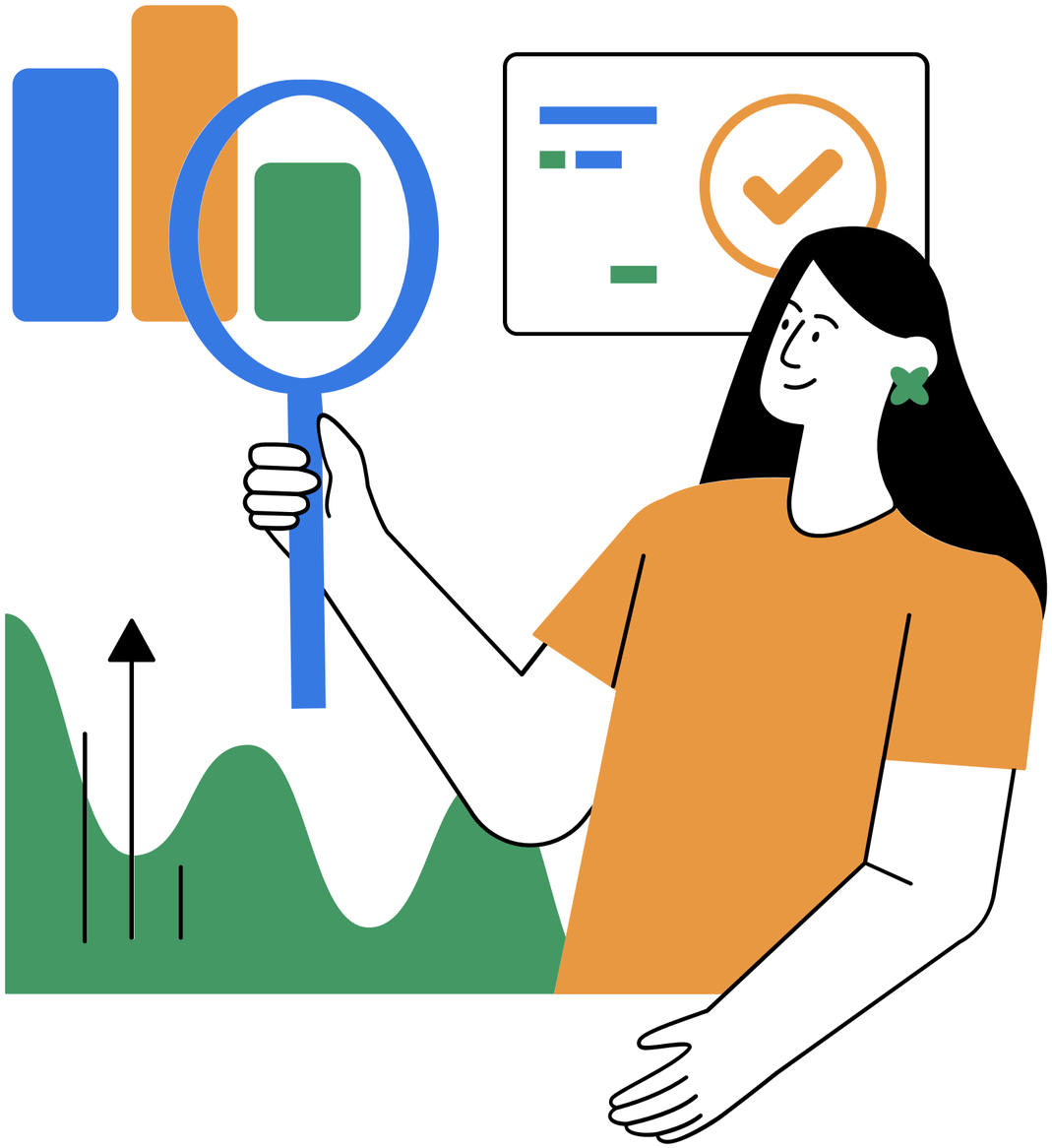
Spotify Virality Study



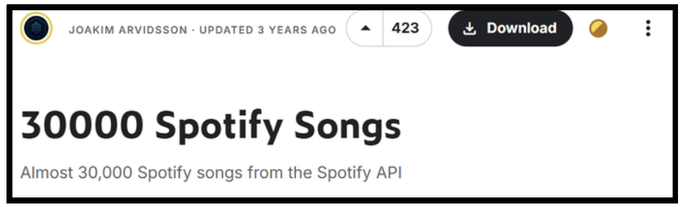
The Question

Can we predict how popular a song will be based purely on its audio DNA?

- **32,833** tracks from the Spotify Web API
- **23** features: danceability, energy, loudness, tempo, valence, and more
- Target: track_popularity (0–100 continuous score)

The Data

Source: Kaggle



→ **32,833** tracks, **23** features

Features:

AUDIO
- danceability
- energy
- acousticness
- liveness
- valence
- loudness
- speechiness
- instrumentalness
- tempo

CATEGORICAL
- playlist_genre
- playlist_subgenre

TEMPORAL
- release_year
- release_month

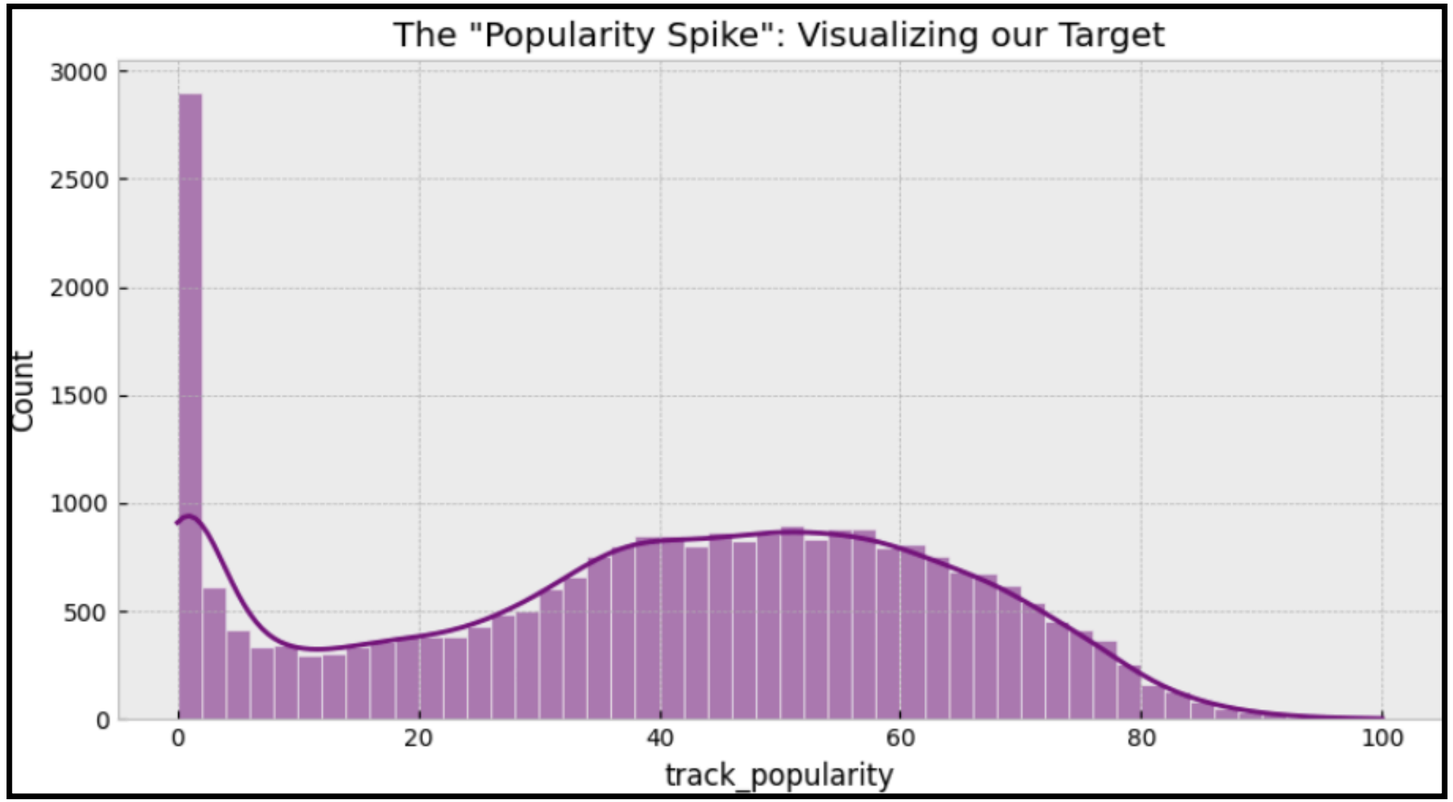
ENGINEERED
- intensity_index
- dance_tempo_ratio

Cleaning the Data

Step	Result
Remove duplicate track IDS	**-3961** entries
Dropped rows with nulls	**-315** entries
Final dataset	**28,557** unique tracks

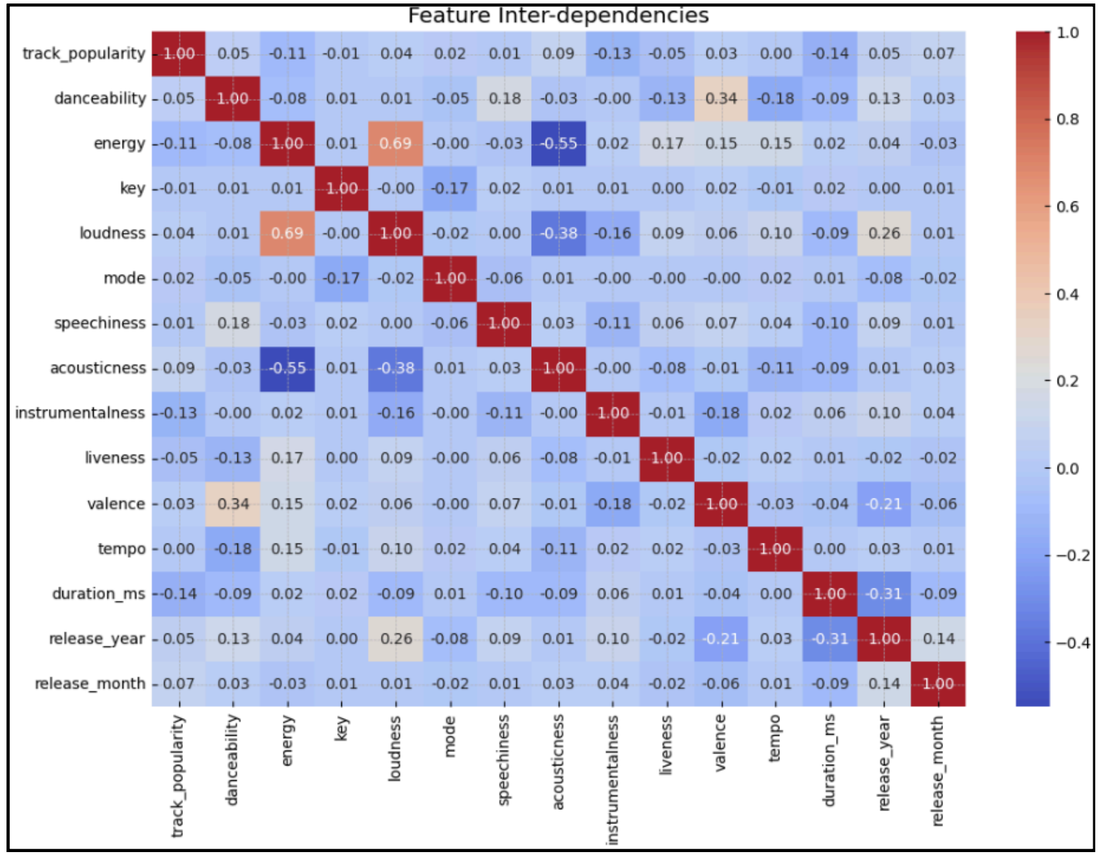
- 800 loudness outliers detected via IQR - retained, handled with RobustScaler
- Release dates converted to datetime → extracted year and month as features
- No imputation needed - dataset was clean post-deduplication

Visualising Data



- Popularity is heavily right-skewed - most tracks cluster near zero
- Only a small fraction of tracks achieve scores above 70
- This reflects the reality of streaming: **a few tracks capture most attention**
- This skew would later make "hit prediction" a difficult classification problem

Do Audio Features Predict Popularity?



Barely.

- Highest correlation with popularity: instrumentalness at -0.13
- danceability, energy, valence, tempo - all under 0.11
- No single audio feature is a meaningful linear predictor

However, the heatmap does show:

- **energy ↔ loudness**: 0.69 - these features move together
- **acousticness ↔ energy**: -0.55 - acoustic tracks are distinctly quieter and calmer
- **valence ↔ danceability**: 0.34 -happier songs tend to be more danceable

Modelling Approach

We built a unified preprocessing pipeline and tested five models in sequence:

RobustScaler + OneHotEncoder → Model

Stage	Model	Purpose
Baseline	Linear Regression	Establish floor
Baseline	Ridge Regression	Check for overfitting
Non-linear	Polynomial Ridge (degree 2)	Capture curves
Ensemble	Random Forest	Bagging
Ensemble	XGBoost	Gradient boosting
Final	Stacking Ensemble	Combine best models

*** Categorical features (genre, subgenre) encoded via OneHotEncoder and fed into every model.**

Results

	Model	RMSE
4	XGBoost	22.351500
2	Polynomial Ridge (Degree 2)	22.599800
3	Random Forest	22.724000
1	Ridge Regression	22.995200
0	Linear Regression	22.995300

```
Nuclear Stacking Ensemble evaluation complete. RMSE: 22.3446
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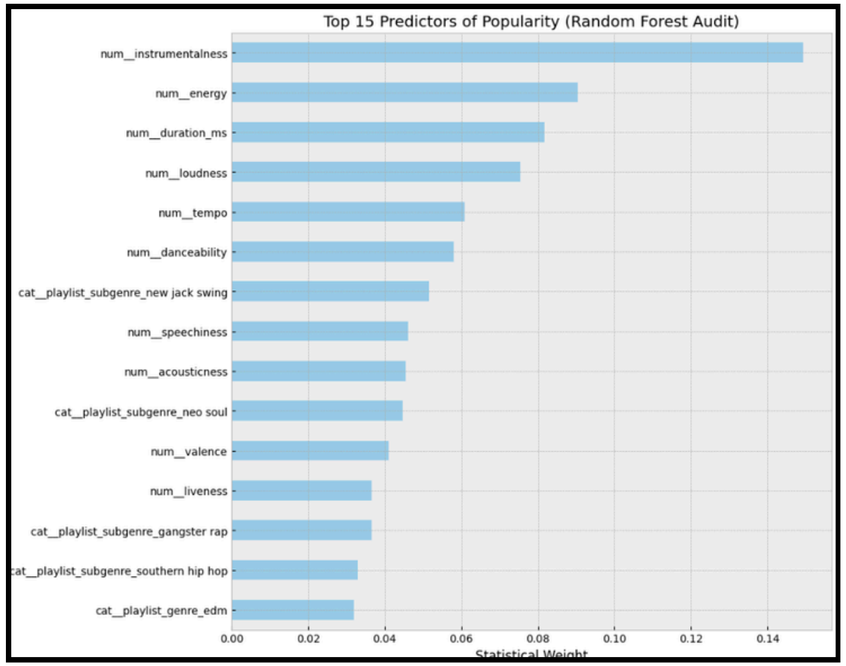
- Ensemble models outperform linear baselines
- XGBoost improvement over Random Forest is statistically significant ($p = 0.026$)
- But the overall ceiling is modest, as the best RMSE of 22.3 on a 0–100 scale means meaningful prediction error remains

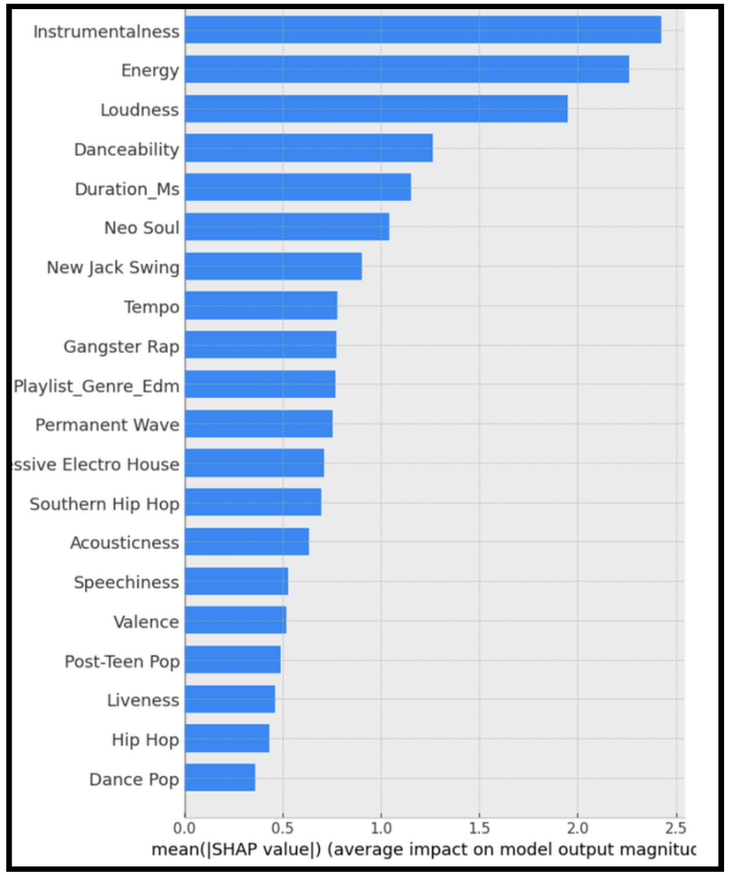
What's Driving Popularity ?

RF Feature Importance Chart



Finding Similarities

SHAP Chart



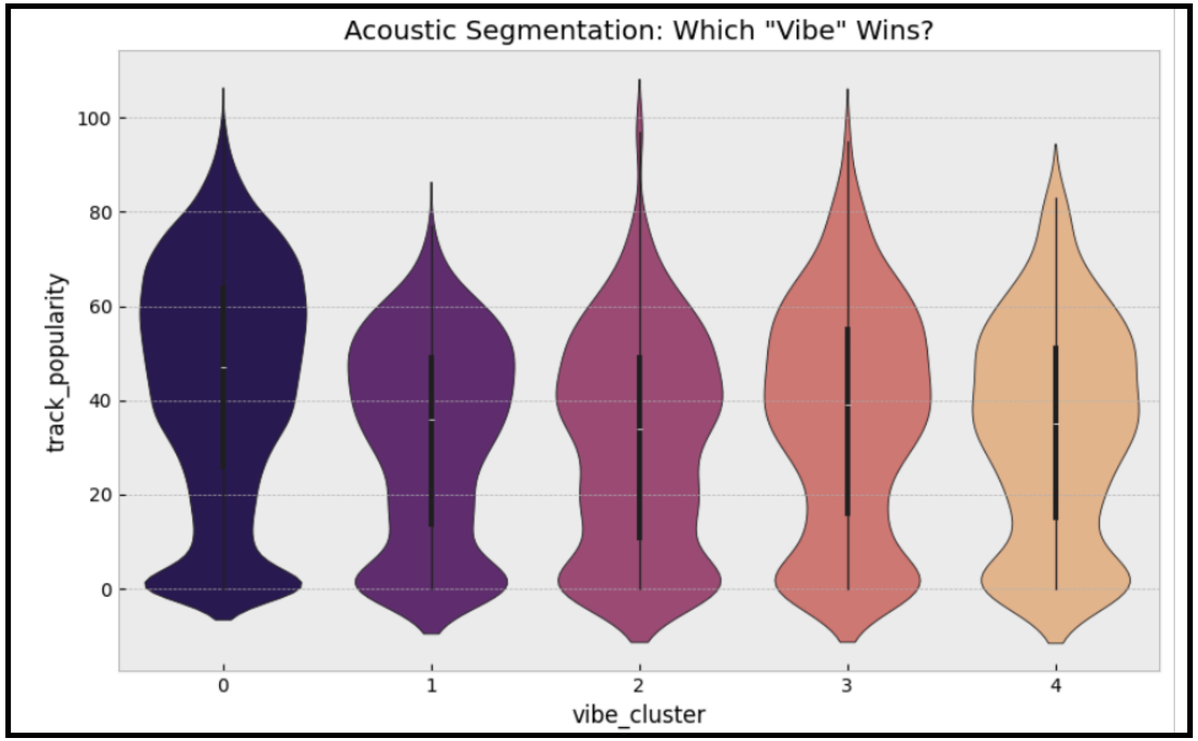
Audio features are the primary drivers:

- **Instrumentalness** — #1 in both RF and SHAP by a wide margin
- **Energy** — consistently #2
- **Loudness, Danceability, Duration_ms** — top 5 in both

Genre/subgenre plays a real but secondary role:

- Neo Soul, New Jack Swing, Gangster Rap, and EDM appear in positions 6–13 on SHAP
- They add signal, but don't dominate

Vocal Tracks Win



Cluster	Profile	Avg Popularity
0 — Mainstream Vocal	Low instrumentalness, moderate energy	**44.1** ✓
3 — Energetic Vocal	Low-mid instrumentalness, high energy	36.4
4 — Mixed Energy	Mid instrumentalness, high energy	33.9
2 — High-Energy Instrumental	High instrumentalness, highest energy	31.2
1 — Instrumental/Ambient	Dominant instrumentalness, quietest	32.1

- Cluster 1 (Instrumental/Ambient) has the highest instrumentalness, which SHAP said is the #1 predictor, yet it has a mean popularity of only 32.1, the second lowest

- Cluster 0 (Mainstream Vocal, lowest instrumentalness) has a mean popularity of 44.1, the highest

This makes sense directionally — SHAP found that low instrumentalness drives popularity up, not high. Thus, **vocal tracks dominate on Spotify.**

Hit Classifier (popularity > 70):

- Accuracy: 87% - but recall on actual hits is only 2%
- The model almost always predicts "not a hit" because hits are genuinely rare

Conclusions

Can audio features predict Spotify popularity?

Yes, but not in the way we expected.

- **Linear correlation was misleading:** No single audio feature correlated with popularity above 0.13. The signal looked weak.
- **Non-linear models revealed the truth:** Tree-based methods (Random Forest, XGBoost) surfaced that instrumentalness, energy, and loudness are genuine drivers, just not in a linear way
- **The direction matters:** Low instrumentalness, high energy, loud production pushes popularity up. Vocal tracks consistently outperform instrumental ones.
- **Genre adds signal but doesn't dominate:** Subgenres appear in the middle of feature importance rankings, not at the top
- **Best model: Stacking Ensemble:** RMSE of 22.3 on a 0–100 scale. Meaningful but not perfect — popularity has a human and algorithmic component that no model fully captures.

Next Steps

Limitation	What We'd Do Next
Hit classifier fails on class imbalance	Apply SMOTE or adjust class weights — only 13% of tracks are "hits"
Clusters unnamed in original notebook	Profile each cluster's genre composition, not just audio features
No artist-level context	Add follower count, prior hit rate, label affiliation
Static snapshot dataset	Retrain on rolling Spotify API data to capture trends over time
Popularity has a recency bias	Model popularity decay — newer tracks score higher temporarily
Linear RFE misled feature selection	Re-run RFE with a tree-based estimator for more reliable results

Thank You

- Mauurya
- Mayank